BHANU PRAKASH VELAGA

Tampa, FL — velagabhanuprakash@gmail.com — (813) 393-9798

SUMMARY

Software Developer with experience designing, building, and optimizing backend applications and RESTful APIs using **Java, Spring Boot, Python, and SQL**. Strong foundation in **object-oriented programming, data structures, and SDLC**, with hands-on experience in debugging, unit and integration testing, CI/CD pipelines, and performance optimization. Proven ability to deliver scalable, reliable software in Agile environments.

EDUCATION

University of South Florida, Tampa, FL — **08/2024 – 12/2025**
Master of Science in Computer Science — GPA: 3.95
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, Machine Learning, Artificial Intelligence

SKILLS

Programming Languages: Java, Python, C, C++, JavaScript, SQL
Backend & APIs: Spring Boot, RESTful Services, Backend Development, Application Development
Databases & Cloud: MySQL, Google Cloud Platform
Tools & Practices: Git, Docker, Linux, CI/CD Pipelines, Automated Testing, Debugging, Performance Optimization, Agile/Scrum, SDLC, OOP, Data Structures, System Design

PROFESSIONAL EXPERIENCE

Conscendo Technologies — **08/2025 – 12/2025**
Software Engineer Intern — Dover, DE

- Analysed backend services and RESTful APIs using **Python and Django/Flask**, improving application performance by **25%**
- Designed and implemented validation logic and automated workflows in Python, reducing data entry errors by **15%**
- Optimized processing logic and SQL queries, increasing system throughput by **30%**
- Created unit and integration tests and supported CI/CD pipelines, achieving **95%+** test coverage for production releases

ProXforce Technologies — **05/2021 – 12/2023**
Software Engineer — Hyderabad, India

- Directed backend services for enterprise-scale applications serving **50,000+** users using **Java, SQL, and REST APIs**
- Integrated **4+ external systems** via REST APIs, securely processing **5,000+** records per day
- Automated internal processing workflows, eliminating **200+ hours/month** of manual effort
- Improved database performance by **35%** through SQL query optimization and performance tuning

RESEARCH & PUBLICATIONS

Driver Drowsiness Detection System using Machine Learning — **12/2022 – 04/2023**
Machine Learning, Computer Vision

- Decreased model inference time by **12%** through composed image preprocessing and efficient resource allocation while maintaining a **93%** detection rate
- Implemented adaptive alert mechanisms for continuous monitoring and fatigue mitigation
- Deployed the trained model into a real-time application pipeline handling image preprocessing, inference execution, and alert generation under live conditions

Gesture-Controlled Virtual Mouse — **08/2022 – 11/2022**
Computer Vision, Human-Computer Interaction

- Created image preprocessing and inference steps, reducing average inference latency by **80ms** and improving system responsiveness
- Evaluated model performance under real-world conditions and refined preprocessing and inference steps to ensure stability and low latency
- Engineered an adaptive image enhancement pipeline leveraging histogram equalization and wavelet transforms, which decreased image noise by 30% across variable lighting conditions, and reduced error rate by 10%.

LEADERSHIP & AFFILIATIONS

President, Student Governance (SG), VVIT
President, ACM Student Chapter, VVIT
Events Lead, Student Governance (SG), VVIT